COMP3613 Assignment 2

Student Accommodation Project

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COMP3613 Assignment 2 Report

1. Cover Page

D'mari Williams ID: 816043902 **COMP3613 Assignment 2 Student Accommodation Project**

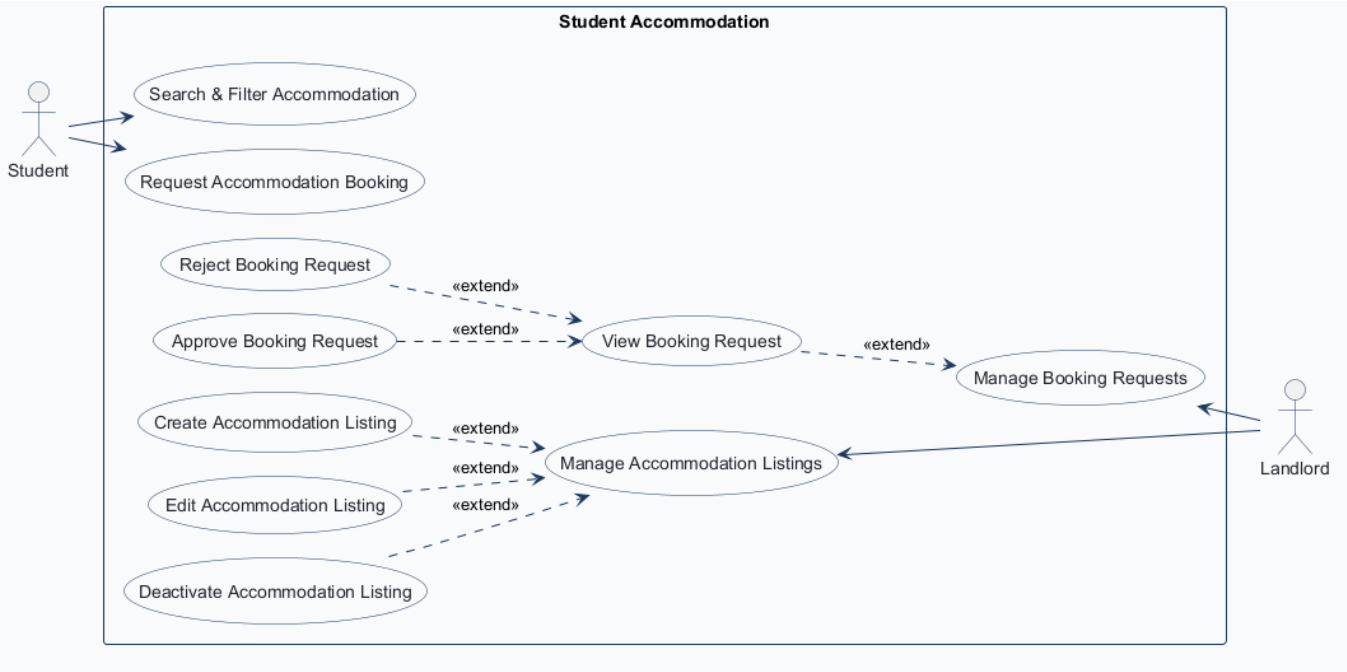
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2. Project Description

The project is an Airbnb-like student accommodation marketplace. Students can search and filter nearby accommodation and submit booking requests, while landlords can manage accommodation listings and booking requests. This report continues the Student Accommodation project documented in Assignment 1.

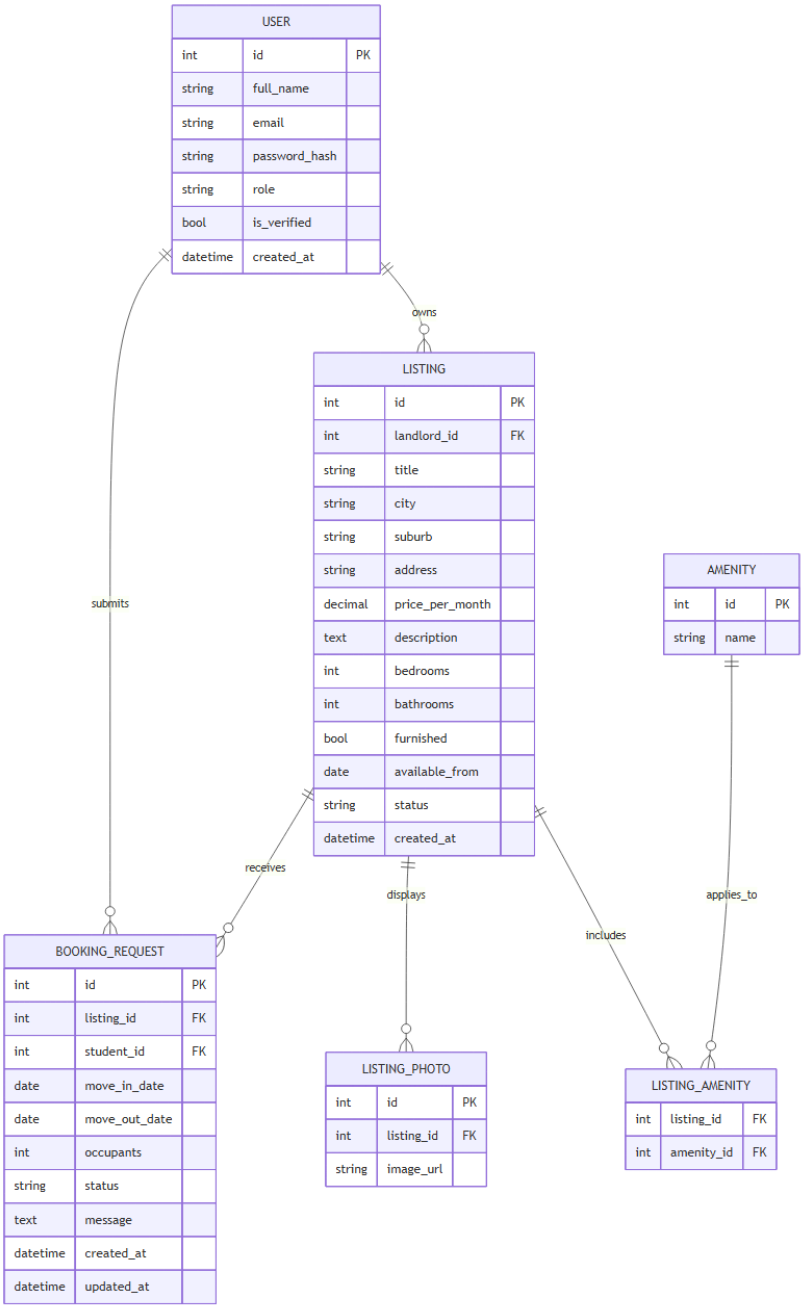
3. Use Case Diagram

The existing Assignment 1 use-case diagram shows the Student and Landlord actors and the project's search, booking, listing-management, and booking-request workflows.



4. Entity Relationship Diagram

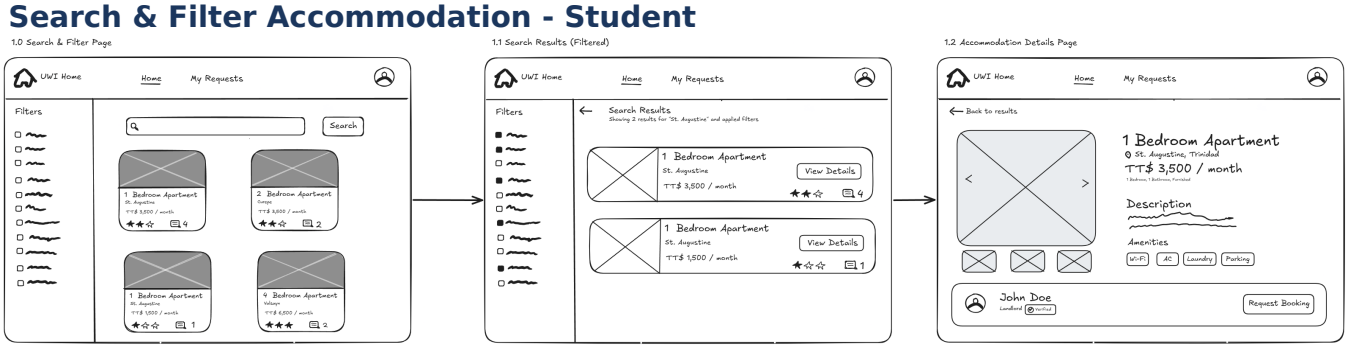
The following ERD source is taken from the Assignment 1 report.



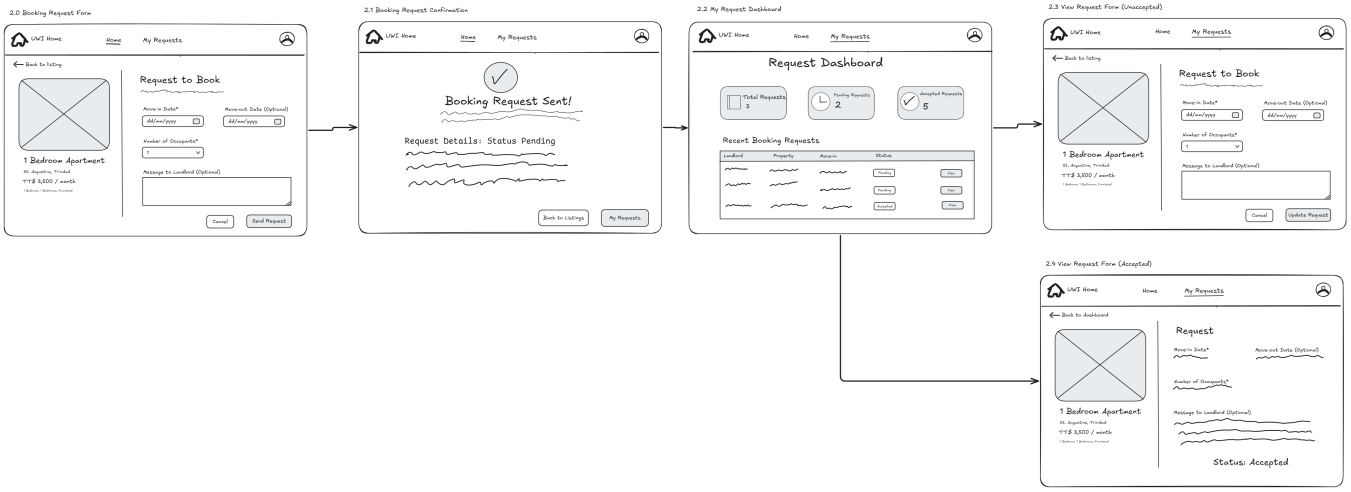
5. Application Wireframes

The following are the existing Assignment 1 wireframes.

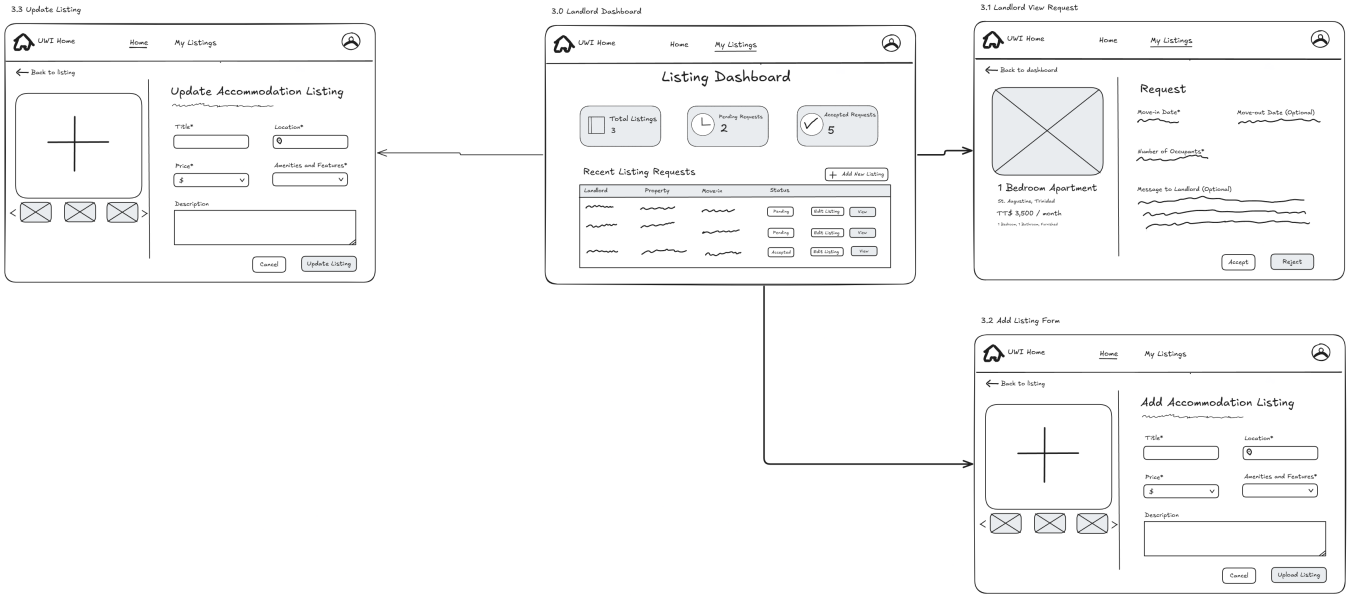
Search & Filter Accommodation - Student



Request Accommodation Booking - Student



Manage Listings & Booking Requests - Landlord



6. Test Planning

User Acceptance Test Plan

The UAT workflows below are based on the Assignment 1 wireframes. All three planned workflows were manually executed and passed.

Test Case	Pre-conditions	Test Steps	Test Criteria / Expected Result	Success
Search & Filter Accommodation — Student	Application is available; student is signed in; at least two listings with different locations or attributes are available	Open accommodation search; enter a location and select filters; submit; review results and open listing details.	Results match the search and selected filters. Details show the accommodation, images, amenities, and landlord, with a way back to results.	Pass
Request Accommodation Booking — Student	Student is signed in; an available listing can be opened and supports a valid requested date.	Open listing details and choose Request Booking; enter move-in date, optional move-out date, occupant count, and optional message; send request; open My Requests	Confirmation is shown. The student's dashboard shows the new Pending request and its details. Pending requests have update/cancel controls; accepted requests are read-only.	Pass
Manage Listings & Booking Requests — Landlord	Landlord is signed in; account has a listing and a request for one of its listings.	Open the landlord dashboard; review listing and request information; open a request and accept or reject it; return to the dashboard and open Add Listing or Edit Listing; submit listing information.	Dashboard shows listings and request statuses. A pending request can be accepted or rejected and its updated status is reflected. The landlord can submit a new listing or save listing changes.	Pass

Unit Test Plan

Unit tests test methods and functions in isolation. Their repository dependencies are mocked, and they have no database dependency.

Unit Test	Module Tested	Expected Output
test_search_normalizes_appartment_to_apartment	ListingService.search_listings	Normalizes the misspelled search phrase and passes the normalized location and default filters to the repository.
test_search_extracts_exact_bedroom_count	ListingService.search_listings	Removes the exact-bedroom phrase from the location and passes the exact bedroom count to the repository.
test_search_passes_other_filters_to_repository	ListingService.search_listings	Forwards the normalized location and supplied price, bedroom, bathroom, furnishing, and availability filters.
test_listing_details_returns_none_when_listing_missing	ListingService.listing_details	Returns None when the repository cannot find an available listing.
test_listing_details_returns_related_data	ListingService.listing_details	Returns the listing, landlord, photos, and amenities, retrieving related records through the repository.
test_landlord_cannot_submit_booking_request	BookingRequestService.require_booking_student	Raises PermissionError for a landlord account.
test_missing_user_cannot_submit_booking_request	BookingRequestService.require_booking_student	Raises PermissionError when the user does not exist.
test_move_in_before_available_date_is_invalid	BookingRequestService._validate_request	Raises ValueError when the requested move-in date precedes the listing's available date.
test_move_out_must_be_after_move_in	BookingRequestService._validate_request	Raises ValueError when the move-out date is not after the move-in date.
test_occupants_must_be_at_least_one	BookingRequestService._validate_request	Raises ValueError when the occupant count is below one.
test_valid_booking_request_is_created	BookingRequestService.create_booking_request	Creates and returns a pending request with the supplied student, listing, occupant, and date data.
test_non_pending_request_cannot_be_updated	BookingRequestService.update_booking_request	Raises ValueError when attempting to update a non-pending request.
test_pending_request_can_be_cancelled	BookingRequestService.cancel_booking_request	Cancels a pending request through the repository and returns the request.
test_non_landlord_is_rejected	LandlordManagementService.require_landlord	Raises PermissionError for a non-landlord account.
test_missing_user_is_rejected	LandlordManagementService.require_landlord	Raises PermissionError when the account does not exist.

Unit Test	Module Tested	Expected Output
test_dashboard_calculates_totals	LandlordManagementService.dashboard	Returns total listings, pending requests, and accepted requests counts from repository data.
test_update_listing_fails_if_listing_not_owned	LandlordManagementService.update_listing	Raises LookupError when the landlord does not own the requested listing.
test_invalid_booking_decision_is_rejected	LandlordManagementService.decide_booking_request	Raises ValueError for an unsupported booking decision.
test_non_pending_booking_request_cannot_be_decided	LandlordManagementService.decide_booking_request	Raises ValueError when attempting to decide a request that is not pending.
test_valid_pending_request_can_be_accepted	LandlordManagementService.decide_booking_request	Accepts a valid pending request and asks the repository to persist the accepted status.

Integration Test Plan

Integration tests use a per-test, in-memory SQLite database. They use actual models, repositories, services, and database sessions to verify persistence and interactions between those components and the database.

Test	Dependencies	Description
test_user_is_saved_and_retrieved_from_database	In-memory SQLite test database, User, UserRepository	Saves a user, commits and refreshes it, then retrieves it by ID and verifies username and email.
test_listing_is_saved_and_found_by_search	In-memory SQLite test database, User, Listing, ListingRepository, ListingService	Persists users and a listing, searches through the service/repository, and verifies the matching listing is returned.
test_booking_request_is_created_and_persisted	In-memory SQLite test database, user/listing/booking request models and repositories, BookingRequestService	Creates a booking request through the service and verifies the request and its student, listing, message, and pending status are persisted.
test_booking_request_can_be_cancelled_and_change_persists	In-memory SQLite test database, user/listing/booking request models and repositories, BookingRequestService	Creates and cancels a request through the service, then reloads it and verifies its status is cancelled.
test_student_dashboard_reads_booking_requests_from_database	In-memory SQLite test database, user/listing/booking request models and repositories, BookingRequestService	Creates two requests and verifies the student dashboard totals, statuses, and rows reflect database records.
test_inactive_listing_is_not_returned_in_student_search	In-memory SQLite test database, user/listing models, ListingRepository, ListingService	Persists an inactive listing and verifies student search does not return it.

7. Test Implementation

Repository Link

BossDNW/comp3613-a1

Unit Tests

The existing unit test files contain 20 passing tests:

Pytest file	Tests	Coverage summary
tests/unit/test_listing_service.py	5	Listing search normalization and filter forwarding; listing details and related data.
tests/unit/test_booking_request_service.py	8	Student authorization, request validation, creation, update restrictions, and cancellation.
tests/unit/test_landlord_management_service.py	7	Landlord authorization, dashboard counts, listing ownership, and booking decision validation and acceptance.

Integration Tests

tests/integration/test_accommodation_workflows.py contains 6 passing tests covering user persistence, listing search, booking-request creation and cancellation, student dashboard reads, and exclusion of inactive listings from student search.

API/Postman Tests

The following REST API routes are implemented and documented for Postman testing:

Method and route	Tested outcome
GET /api/listings	200
GET /api/listings/{listing_id}	200
Missing listing	404
Unauthenticated booking request	401
Successful login	303
Authenticated booking request	201
Invalid booking request	422
GET /api/requests	200

The Postman collection was completed and exported to postman/COMP3613_A2.postman_collection.json.

8. Test Reporting

Test area	Result
Unit tests	20 passed
Integration tests	6 passed
Postman collection	17 passed, 0 failed, 0 skipped, 0 errors
User acceptance testing	3/3 planned workflows passed

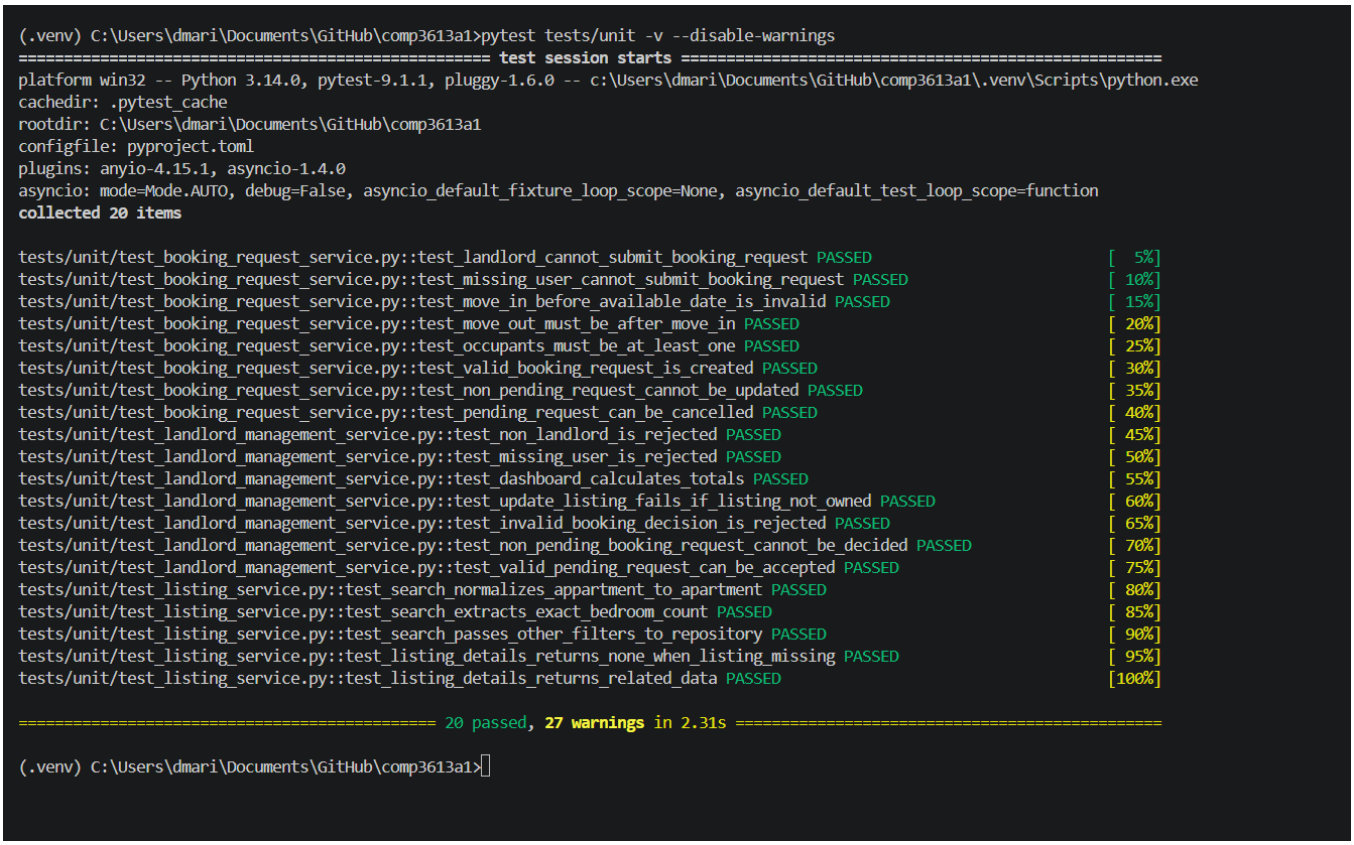
Verification Results

The existing pytest suite was run with python -m pytest -q tests\unit tests\integration: **26 passed** (20 unit and 6 integration).

9. Evidence

The following existing screenshots are included as evidence.

Unit Test Screenshot



Integration Test Screenshot

```
(.venv) C:\Users\dmari\Documents\GitHub\comp3613a1>pytest tests/integration -v --disable-warnings
============================================================ test session starts ============================================================
platform win32 -- Python 3.14.0, pytest-9.1.1, pluggy-1.6.0 -- c:\Users\dmari\Documents\GitHub\comp3613a1\.venv\Scripts\python.exe
cachedir: .pytest_cache
rootdir: C:\Users\dmari\Documents\GitHub\comp3613a1
configfile: pyproject.toml
plugins: anyio-4.15.1, asyncio-1.4.0
asyncio: mode=Mode.AUTO, debug=False, asyncio_default_fixture_loop_scope=None, asyncio_default_test_loop_scope=function
collected 6 items

tests/integration/test_accommodation_workflows.py::test_user_is_saved_and_retrieved_from_database PASSED                    [ 16%]
tests/integration/test_accommodation_workflows.py::test_listing_is_saved_and_found_by_search PASSED                          [ 33%]
tests/integration/test_accommodation_workflows.py::test_booking_request_is_created_and_persisted PASSED                      [ 50%]
tests/integration/test_accommodation_workflows.py::test_booking_request_can_be_cancelled_and_change_persists PASSED          [ 66%]
tests/integration/test_accommodation_workflows.py::test_student_dashboard_reads_booking_requests_from_database PASSED       [ 83%]
tests/integration/test_accommodation_workflows.py::test_inactive_listing_is_not_returned_in_student_search PASSED           [100%]

============================================================ 6 passed, 14 warnings in 2.40s =============================================================

(.venv) C:\Users\dmari\Documents\GitHub\comp3613a1>
```

Postman Collection Run Screenshot

